Hide the Zeros: A 10-Minute Place Value Lesson

A free lesson for elementary classrooms, built around Hide Zero Cards, a free digital tool. Grades 3–4. Works for grades 2–5 by changing the Roll range.

Open the tool: **hidezerocards.org** No login, no ads, no student accounts. Works on projectors, smartboards, tablets, and Chromebooks.

What it is

Hide Zero Cards is the digital version of place value arrow cards. You roll or type a number, and it fans out as color-coded cards, one for each digit. Tap **Hide zeros** and the zero cards hide in place, so 3,743 reads as 3, 7, 4, 3 while the class talks about what each digit is worth. Pull the 3 card out and it shows **3,000**, the value that was tucked behind it all along.

You will need

10 minutes. A classroom display, or one device per student. Nothing to print, nothing to install, no accounts.

The lesson

1. **Roll or type a number.** Start with 3,743, or type a number from your current unit.
2. **Hide the zeros.** Ask: what is each digit worth? The 3 is 3,000, the 7 is 700, the 4 is 40, the 3 is 3.
3. **Pull one card out.** Pull the 3 to show 3,000, then the 7 to show 700. The digit was that value all along.
4. **Try 2,609.** Hide the zeros and ask what changed. The zero in the tens place hides, and the 600 card shows in full. Ask: is there no digit there, or is the digit zero?
5. **Mix and rebuild.** Tap Mix to scatter the cards, then have students drag them back into place value order.
6. **Number Forms.** Show the same number in word, expanded, and unit form, plus standard form.

Try these numbers

3,743: hide the zeros, then pull the 3 out for 3,000. **2,609:** what does the zero in the tens place mean? **180,736:** start smaller, then build up. **Roll:** ranges run from up to 100 to up to 1,000,000,000, so set the range to the level you are teaching.

Talk prompts

What is the 3 in 3,743 worth, and how do you know? What changes when we hide the zeros, and what stays the same? If we move the 7 card, what happens to the number?

After the lesson

Tell the builder what happened. Email **donrayxwilliams@gmail.com** with the grade you tried it with, what students understood faster, what confused them, and whether you would use it again next week.